

2(a)	$n = 110 / 0.032$ or $110000 / 32$ or 3440	C1
	$pV = nRT$	C1
	$T = (1.0 \times 10^5 \times 85) / (8.31 \times (110 / 0.032)) = 300 \text{ K}$	A1
2(b)	$E = mc\Delta\theta$ $= 110 \times 0.66 \times 50$	C1
	$= \mathbf{3600 \text{ J}}$	A1
2(c)	Any 3 from: • molecule collides with wall • momentum of molecule changes during collision (with wall) • force on molecule so force on wall • many forces act over surface area of container exerting a pressure	B3
2(d)	$KE \propto T$ $v \propto \sqrt{T}$	C1
	ratio $= \sqrt{(350 / 300)}$ $= \mathbf{1.1}$	A1